

Advanced Chooser Option Pricing Model

Final Project Report

JPM underlying | Week 1 through Week 8

This report documents the complete data, pricing, validation, machine-learning, sensitivity, and application pipeline. The key result is a working dual-pricing tool whose selected volatility model improves test proxy RMSE while retaining BSM as the pricing rule.

Executive Summary

The project developed a Black-Scholes-Merton simple chooser pricing model for JPM and extended it with two machine-learning approaches. Weeks 1 and 2 collected, aligned, cleaned, and engineered JPM, VIX, and Treasury data. Week 3 replicated the paper-matched chooser model with $K=\$150$, $T1=0.5$, and $T2=1.0$. Week 4 established the baseline. Weeks 5 and 6 built and compared ML models. Week 7 added stress tests and live-data infrastructure. Week 8 packaged the full chain into a Streamlit application.

Final pricing route	MAE	RMSE	R ²
Week 4 BSM baseline	3.5698	4.8706	0.9460
Approach 1 RF vol plus BSM	3.1384	4.3203	0.9575
Approach 2 direct price	45.0275	49.9891	-4.6915

Approach 1 reduced proxy RMSE by 11.30% relative to the BSM baseline. Approach 2 failed under the later price regime and is reported as an extrapolation limitation. Public historical chooser transaction prices were not available, so both comparisons use an ex-post forward-volatility BSM proxy. The project does not compare chooser values with JPM stock closes.

Submission Status

- Technical Weeks 1–8 complete
- Final application and model bundle complete
- Public deployment and narrated demo require student account access and recording

1 Project Objective and Contract

A simple chooser option allows the holder to decide at time T_1 whether the contract becomes a European call or a European put with final maturity T_2 . The project uses JPM as the underlying and retains the paper-matched contract so results remain comparable across weeks.

Parameter	Value	Treatment
Strike K	\$150	Required project setting
Choice time T_1	0.5 year	Six-month decision
Maturity T_2	1 year	Required setting
Dividend yield q	2.33%	Paper value retained
Spot, rate, volatility	Date dependent	Week 2 data or current snapshot

Project Chain

Week 1 raw market data -> Week 2 preprocessing and features -> Week 3 chooser BSM -> Week 4 baseline evaluation -> Week 5 ML design -> Week 6 training and interpretation -> Week 7 stress tests and data updater -> Week 8 final tool.

Each stage consumes the previous stage's files. This design prevents the later work from replacing the original market-data contribution or changing the Week 3 pricing economics.

2 Market Data and Feature Engineering

The Week 2 dataset contains 1,760 aligned trading-date observations from 2018-01-02 through 2024-12-30 and 23 cleaned or engineered columns. JPM OHLCV data supplies spot and return information. VIX represents market volatility conditions. Treasury yields provide the risk-free-rate proxy.

Feature group	Examples	Use
Returns and volatility	Daily return, log return, rolling vol	BSM input and ML prediction
Trend and momentum	MA20, MA50, price momentum	Market state
Risk indicators	VIX, VIX return, JPM-VIX correlation	Regime and ML features
Rates	Treasury rate, rate momentum	Discounting and ML features
Trading activity	Intraday range, volume z-score	Short-horizon conditions

Feature construction preserves dates and units. Treasury percentages are divided by 100 before BSM use. Rolling volatility is annualized with 252 trading days. Features use information available on or before each row date. Because each target uses the following 20 trading days, the final 20 rows before both development boundaries are excluded from fitting and selection.

GitHub Actions now reproduces the Week 2 pipeline from the archived Week 1 inputs every Monday and on manual request. The verified run rebuilt the processed data, checked its schema and date range, and uploaded the outputs as an artifact. The project repository is <https://github.com/Frank-Wang-Jc/jpm-chooser-option-pricing>.

3 BSM Chooser Model

Week 3 implemented European call and put functions, the adjusted choice boundary, and the simple chooser decomposition. The analytical price equals a European call expiring at T2 plus an adjusted put component whose strike reflects the remaining period after T1.

Validation test	Result
Put-call parity error	0.000000
Paper choice decisions	10 of 10 matched
Paper payoff calculations	Maximum error approximately 0
Monte Carlo vs closed form	0.823 standard errors
Week 3 vs Week 4 regression	Maximum difference 0.000000

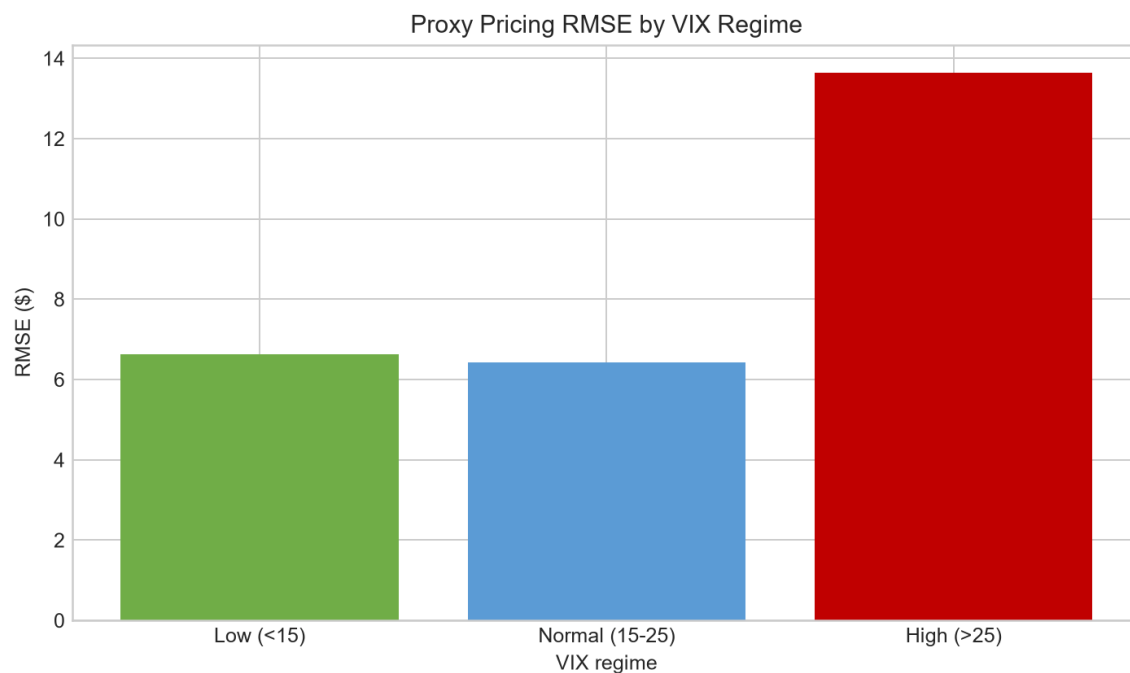
Under the paper inputs, the analytical chooser price was \$29.1299. Under the latest Week 2 baseline, the value was approximately \$82.53 because JPM was substantially above the fixed strike. The higher level is a contract-moneyness result, not a change to the formula.

4 Baseline Evaluation

Week 4 separated computational evidence from market evidence. The historical proxy compares trailing-volatility pricing with a chooser price computed from the next 20 days of realized volatility. The listed market check applies vanilla BSM to delayed JPM call and put quotes.

Benchmark	MAE	RMSE	R ²
Historical chooser proxy	3.7757	8.3567	0.8242
Listed vanilla, Week 2 vol	0.3635	0.4401	0.9845
Listed vanilla, quoted IV	0.1658	0.2201	0.9961

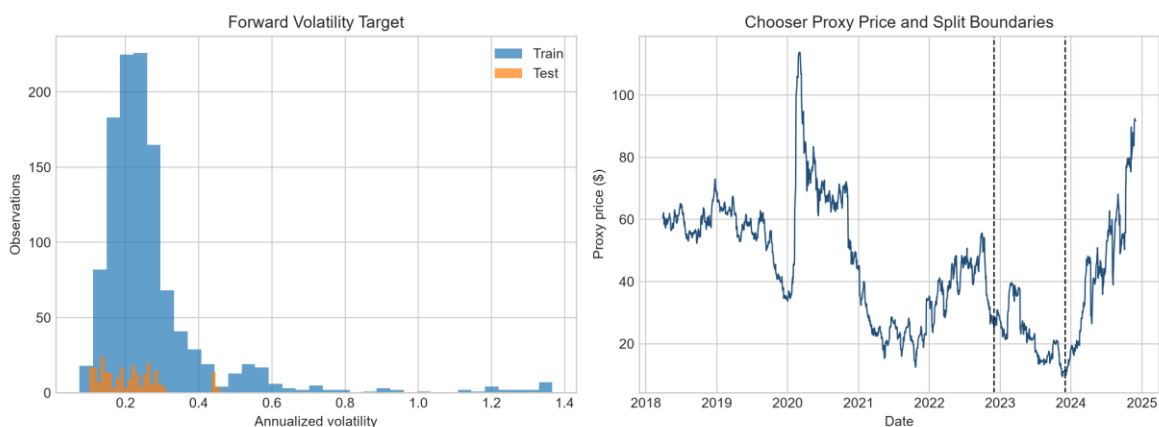
Figure 1 Historical proxy RMSE by VIX regime



5 Machine Learning Design

Approach 1 predicts 20-day forward realized volatility with 17 market features, then passes that forecast through BSM. Approach 2 predicts the forward-volatility chooser proxy directly with 19 observable market features. It excludes the current BSM price and therefore provides the end-to-end supervised benchmark required by the brief.

Figure 2 Target construction and chronological split

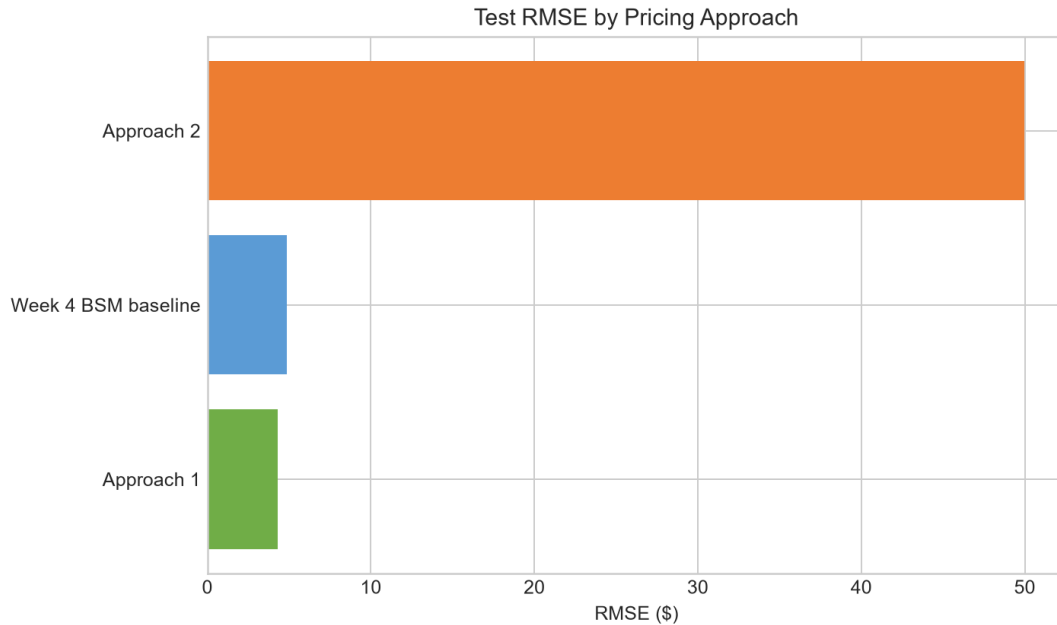


Use	Rows	Period
Train	1,156	2018-03-29 to 2022-10-28
Validation	232	2022-11-29 to 2023-10-31
Purged	40	20 rows before each boundary
Test	252	2023-11-30 to 2024-11-29

6 Training and Test Performance

Linear, Random Forest, XGBoost, and neural-network candidates were tuned with expanding-window cross-validation using a 20-day gap. Validation RMSE selected each route. The final test set was evaluated once afterward. Approach 1 achieved the best price metrics; direct Approach 2 failed under the later price regime.

Figure 3 Held-out proxy RMSE



Volatility forecast	MAE	RMSE
Persistence	0.085584	0.120650
Selected Random Forest	0.072424	0.094069

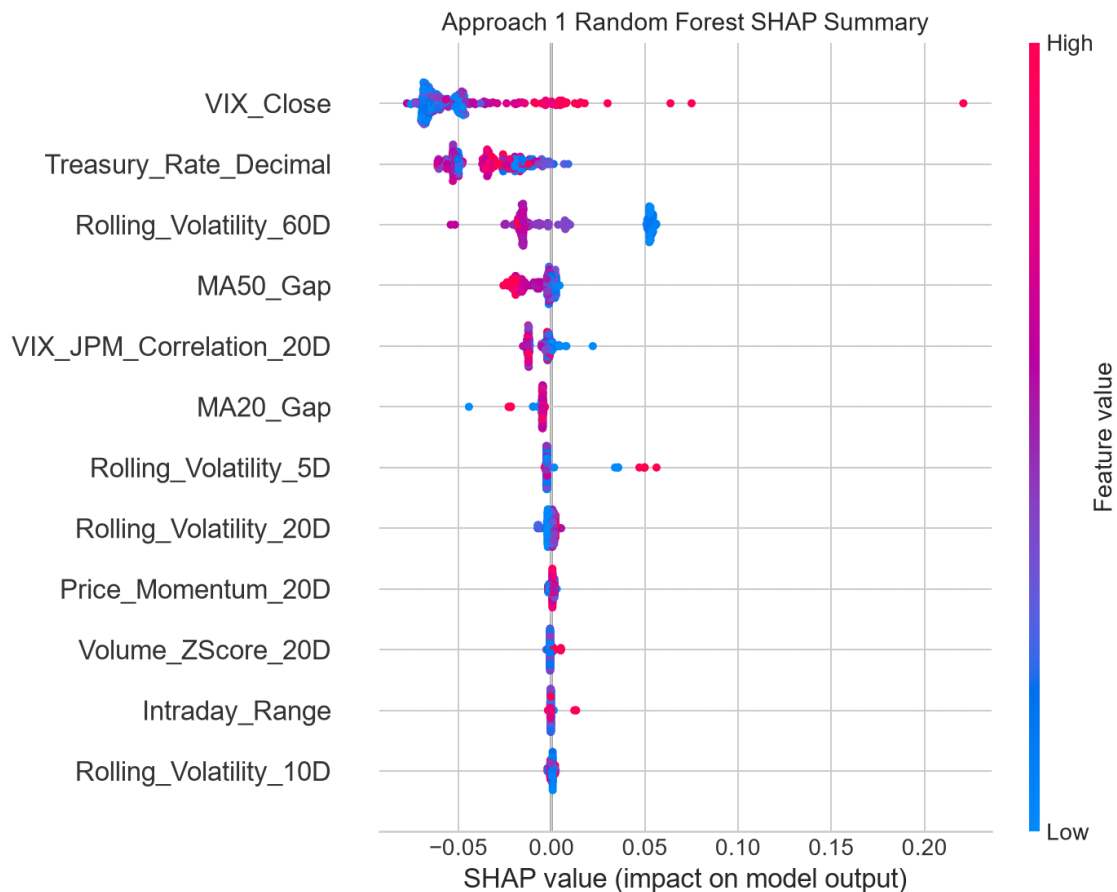
The selected volatility route reduced proxy-price RMSE by 11.30% relative to BSM. The direct model's negative R^2 and RMSE of 49.9891 show that a price-level learner trained below \$152.49 did not extrapolate to the later JPM regime. The project retains this failure as evidence for keeping the financial baseline visible.

Both selected Week 6 estimators are archived as native Python pickle files. Each file was reloaded and checked on all 252 held-out rows; the maximum absolute difference from the saved Week 6 predictions was below 1e-12. Joblib runtime copies and model metadata are bundled with the Week 8 application.

7 Model Interpretation

SHAP and permutation importance were used to inspect the exact selected models. Approach 1 explains the selected Random Forest volatility model. Approach 2 explains the selected direct-price Linear Regression model. The current BSM price is absent from Approach 2 by design.

Figure 4 Approach 1 SHAP feature importance

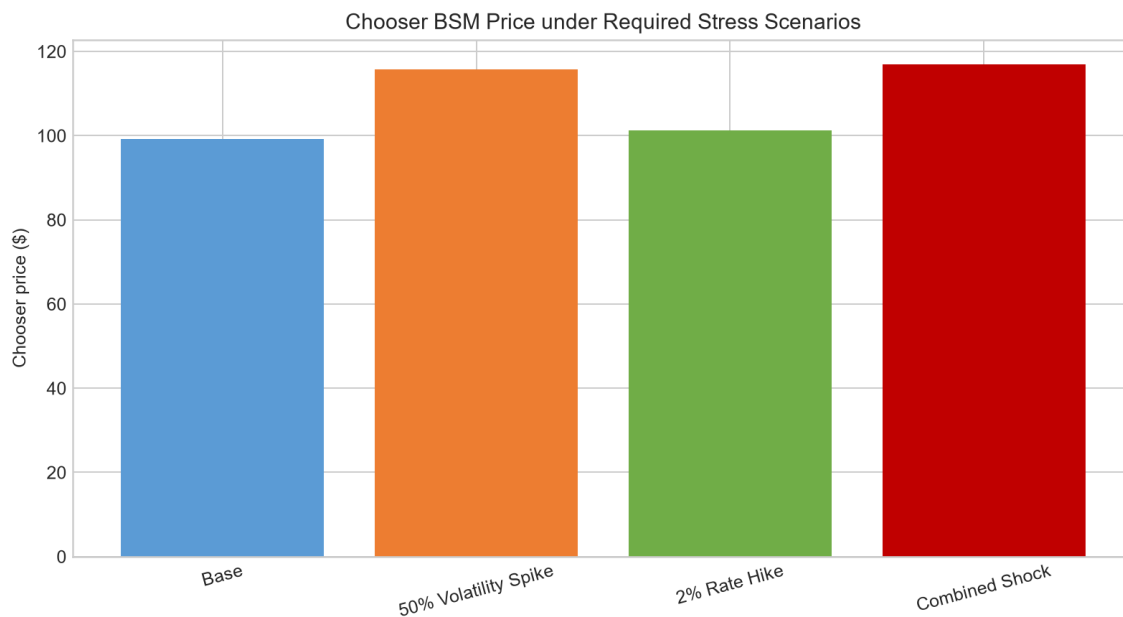


VIX-related features represented 40.80% of total mean absolute SHAP contribution in the selected Approach 1 model and 5.06% in selected Approach 2. These shares describe model reliance within the sample and should not be read as causal effects.

8 Sensitivity and Stress Testing

Week 7 implemented the required 50% volatility spike and absolute 2 percentage-point rate hike. The structural BSM test used the last full Week 5 model row, which had JPM spot \$241.12 and historical volatility 45.72%.

Figure 5 Required Week 7 stress scenarios



Scenario	BSM price	Change
Base	\$99.2617	\$0.0000
50% volatility spike	\$115.6683	+\$16.4066
2% rate hike	\$101.3317	+\$2.0700
Combined	\$116.9177	+\$17.6561

9 Data Update and Final Application

The updater rebuilds all original ML features from JPM OHLCV, Cboe VIX, and FRED DGS10 histories at a common completed date. DGS10 preserves the trained rate feature; DGS1 supplies the one-year pricing rate. Requests have bounded retries and independent source caches. The application refreshes on opening, every 15 minutes while active, and on manual request, retaining a labeled complete snapshot when required data are unavailable.

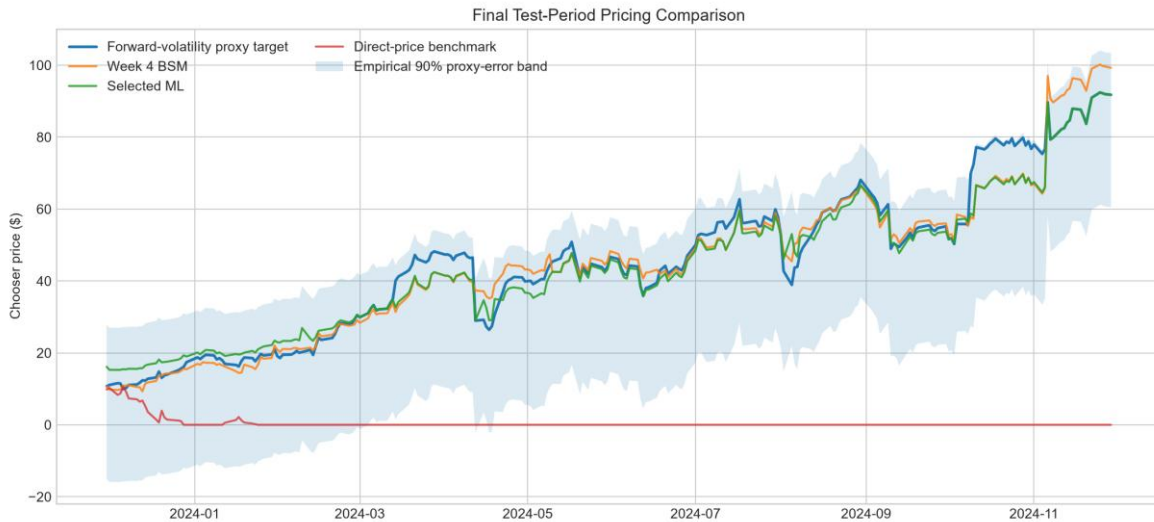
Application tab	Content
Dual pricing	BSM, selected ML-volatility route, direct-price benchmark, Greeks, proxy error band
Sensitivity	Volatility curve, rate curve, required stress scenarios
Model performance	MAE, RMSE, R^2 and held-out price trend
Methods and limits	Target definition, feature date, model boundary and interpretation

The application restricts ML pricing to $K=\$150$, $q=2.33\%$, $T1=0.5$, and $T2=1.0$ because those values define the training target. BSM remains available for other valid contracts. The automated Streamlit test rendered four tabs, seven metrics, and one refresh control without exceptions.

10 Error Margin and Live Extrapolation

The final tool calibrates signed 5th and 95th percentile residuals from 924 purge-aware expanding-window predictions within the training block. It then evaluates the resulting band once on the later 252-row test period. Test coverage was 99.60% and mean width was \$42.8744, so the interval was conservative in the later regime.

Figure 6 Final test-period pricing and proxy-error band



The offline example aligns spot and all engineered features to 2024-12-30. Its \$231.08 spot exceeds the fitted maximum of \$152.49, so the app displays an extrapolation warning. This complete fallback is reproducible without network access. Online updates advance the feature date only when a complete row is available; they do not retrain or alter the original models.

11 Limitations and Controls

Limitation	Control
No public chooser transaction history	Label the target and interval as theoretical proxies
Treasury maturity mismatch in Week 2	Document the proxy and add DGS1 to live updates
Dividend yield not collected	Retain the paper's 2.33% assumption
Tree-model extrapolation	Keep BSM visible and flag values outside training ranges
Different source publication dates	Rebuild every feature at a common completed date and label cached-source use
No reproducible sentiment series	Record the data gap; do not relabel VIX

These controls keep the final interface honest about what the evidence supports. The tool is a research prototype and does not provide trading or investment advice.

Data and Software Sources

- Project brief and supplied Week 1–3 files
- Week 2 processed JPM, VIX, and Treasury dataset
- Cboe delayed listed JPM option quotes and VIX history
- FRED DGS10 and DGS1 series
- Python, NumPy, pandas, SciPy, scikit-learn, XGBoost, SHAP, and Streamlit

12 Conclusion and Submission Checklist

The project now contains a continuous and reproducible Week 1–8 pipeline. The Week 3 chooser model was preserved, Week 4 established a defensible baseline, and the selected Week 6 volatility model improved test proxy RMSE. Week 7 verified the required shocks and the data updater. Week 8 packaged the pricing routes, out-of-fold calibrated error band, performance evidence, and limitations in a tested application.

Submission item	Status
Code, executed notebooks, GitHub repository, and Actions	Complete; repository integrity, Streamlit smoke tests, and scheduled preprocessing passed
Parameter and model files	Complete
Validation, benchmark, sensitivity, and weekly reports	Complete
Final 10–15 page report	Complete
Final presentation	Complete
Public Streamlit URL	Student account action
Narrated 5–10 minute MP4	Student recording action

Recommended Demonstration Flow

Open the app, confirm the source dates, explain the default contract, compare BSM and ML prices, show the proxy-error interval, run the required stress scenarios, review the test metrics, and finish with the target and extrapolation limitations. The Week 8 folder contains a timed script and recording checklist.